

Irulappan T

 Phone  Gmail  linkedin  github  Portfolio

Profile

Creative Python Developer with a deep interest in artificial intelligence and modern web technologies. Skilled in building end-to-end AI-driven applications, integrating RAG pipelines, and developing scalable full-stack solutions. Experienced with Python, Flask, Next.js, and cloud tools like AWS, with a strong focus on automation, performance, and delivering impactful real-world solutions.

Experience

- Clap Cut Universe** **Mar 2024 – Present**
Python & Full-Stack Developer *Madurai, India*
 - Developing and maintaining full-stack web applications using Flask, React.js, and SQL.
 - Integrated data pipelines and managed model lifecycle from experimentation to deployment on AWS (EC2, RDS).
 - Built RESTful APIs and automation tools for seamless backend–frontend interaction.
 - Collaborated with AI and web teams to deploy ML models into production environments.
- S3 Technologies** **Sep 2023 – Feb 2024**
Python Developer (Machine Learning) *Madurai, India*
 - Built and trained machine learning and deep learning models for real-world business use cases.
 - Architected end-to-end machine learning pipelines integrating TensorFlow and PyTorch frameworks for production deployment.
 - Developed Flask-based web applications and APIs for model serving and performance monitoring.
 - Enhanced system scalability and reliability through modular code design and AWS integration.
- Queen Mira International School** **Freelance**
Data Analyst *Madurai, India*
 - Conducted in-depth data analysis to identify academic and operational trends and insights.
 - Designed interactive Power BI dashboards to visualize performance and key indicators.
 - Automated reporting workflows and improved decision-making using Python and SQL.

Projects

- Paddy Disease Identification using Deep Learning** **2023**
 - Developed a convolutional neural network (CNN)-based deep learning model for paddy leaf disease classification.
 - Performed image preprocessing using augmentation, normalization, and segmentation to enhance model accuracy.
 - Utilized transfer learning with EfficientNetB0 to achieve high precision in disease detection and classification.
 - Implemented a visualization dashboard for model predictions and performance tracking.
- RAG-Based LLM PDF Query System** **2024**
 - Created a RAG-based LLM system enabling users to upload PDF files, process their content, and retrieve context-aware answers via natural language queries.
 - Built using LangChain, Chroma, and Ollama for embedding generation and conversational retrieval.
 - Designed an interactive chat interface for real-time, document-driven AI assistance.
- AI-Based Retail Sales Forecasting System** **2024**
 - Designed an intelligent forecasting model using time-series analysis and LSTM networks to predict future retail sales.
 - Integrated external factors such as seasonality and promotions for improved forecasting accuracy.
 - Deployed the model with a Flask API and interactive dashboard for visualization and decision support.
- Daily Staff Report System (Flask Web App)** **2025**
 - Built a multi-role Flask web application for daily staff reporting, task tracking, and performance analysis.
 - Integrated AWS RDS for secure and scalable cloud-based data storage.
 - Implemented role-based dashboards (Admin, Academic, and MD) with analytics and automated summary reports.
- School Management System (Next.js + Prisma + MySQL)** **Ongoing — 2025**
 - Developing a full-stack school management platform using Next.js, Prisma ORM, and MySQL for seamless data management.
 - Implementing authentication, attendance tracking, grade management, and fee monitoring modules.
 - Designing an intuitive UI with responsive layouts and API-driven backend for real-time performance and scalability.

Technical Skills

Programming: Python, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks: Flask, Django, React.js, Next.js

AI / ML / DL: TensorFlow, PyTorch, Scikit-learn, LangChain

Visualization: Power BI, Matplotlib, Seaborn

Database: MySQL, MongoDB

Development / Tools: Git, Docker, Jupyter, VS Code, Anaconda, Linux, AWS

Education

The American College

Master's Degree in Data Science

Mannar Thirumalai Naicker College

Bachelor's Degree in Computer Applications

2021 – 2023

Madurai, India

2018 – 2021

Madurai, India